

ShopLane E-commerce Website

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The issues faced:

During the development and testing of this website, the main issue I faced was finding a third-party API that I could use to verify credit card details. All require payment and also that one must be registered as a business in order to access the APIs.

The other issue I encountered was the encryption of user data like their CVV and card numbers. I looked through many APIs and finally settled on using Crypto which is a Javascript library.

The other issue was the storage of images in MongoDB. Storing images into MongoDB was a bit challenging as it required them to be stored as a buffer. Retrieving images then becomes an issue especially if one wants to use base64. For this, I settled on using links that reduce the space and also ensure clear images are displayed on the landing pages.

Lessons Learned:

Using express to make endpoints

Using Axios to create REST APIs to call endpoints

Using NodeJs to create a server

Using Mongoose to create schemas and also perform CRUD operations into mongoDB.

Design UI with CSS and also use React for inline use of JavaScript, CSS, and HTML.

Encryption of data using CrptoJs

Tokenization for authentication of users

Hashing of passwords using bcrypt

Remaining bugs:

Dynamic review of a user who has already input their card details before.

The amount of time required to display items before they are automatically deleted

Correct authentication and security of routes from the server

Git Link: <https://github.com/jyothiswaroop-sreenivasulureddy/comp5130/tree/final>